Cambridge (CIE) A Level Computer Science

Your notes

Compression

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- Compression
- Compression Techniques

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Your notes

The need for compression

What is compression?

- Compression is **reducing the the size of a file** so that it **takes up less space on secondary storage**
- There are scenarios where compression may be needed, such as:
  - **Maximise the amount of data you can store** on a digital device such as a mobile phone or tablet
  - **Minimise the transfer time** of data being uploaded, downloaded or streamed across a network such as the Internet
- Compression can be achieved using two methods, **lossy** and **lossless**

Lossy vs lossless

What is lossy compression?

- Lossy compression is when **data is lost** in order to **reduce the size** on secondary storage
- Lossy compression is **irreversible**
- Lossy can **greatly reduce the size** of a file but at the **expense of losing quality**
- Lossy is only suitable for data **where reducing quality is acceptable**, for example images, video and sound
- In photographs, lossy compression **will try to group similar colours together**, reducing the amount of colours in the image without compromising the overall quality of the image

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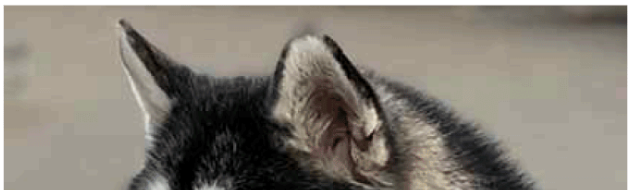
ORIGINAL JPG
824KB

50% LOSSY COMPRESSION
76KB



80% LOSSY COMPRESSION
38KB

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- In the images above, lossy compression is applied to a photograph and **dramatically reduces the file size**
- **Data has been removed** and the **overall quality has been reduced**, however it is acceptable as it is difficult to visually see a difference
- Lossy compressed photographs **take up less storage space** which means you can store more and they are quicker to share across a network

What is lossless compression?

- Lossless compression is when data is **encoded** in order to **reduce the size on secondary storage**
- Lossless compression is **reversible**, the file **can be returned to its original state**
- Lossless can reduce the size of a file but not as dramatically as lossy
- Lossless can be used on all data but is more suitable for data where **a loss in quality is unacceptable**, for example documents
- In a document, lossless compression uses algorithms to analyse the contents looking for **patterns** and **repetition**

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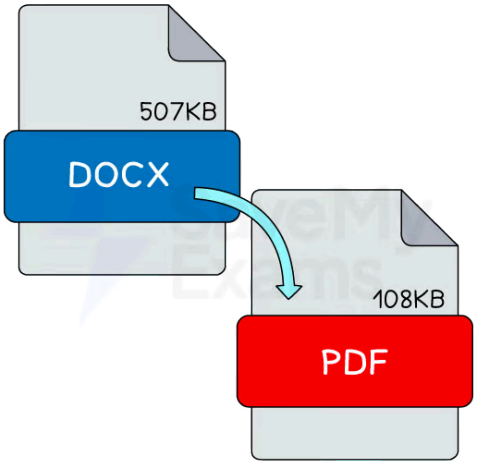

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- In the image above, lossless compression is **automatically applied** to document formats such as **DOCX** and **PDF** with a different rate of success
- When you open a lossless compressed document the **decompression process reverses the algorithms** and returns the data back to its **original state**
- Lossless compressed documents **take up less storage space** which means you can store more and they are quicker to share across a network

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File compression

MPEG-3 (MP3) - Audio compression

- MP3 uses **audio compression** to reduce file size
- A typical music file can be reduced by **up to 90%** (e.g. 80 MB to 8 MB)
- MP3 files are used on:
  - **MP3 players**
  - **Computers**
  - **Smart phones**
- Music can be:
  - **Downloaded or streamed** from the internet
  - **Converted from a CD** to MP3 format
- MP3 quality isn't as high as the original CD, but it's **good enough for most people**
- MP3 uses **perceptual music shaping** to remove sounds we don't notice:
  - Frequencies **outside human hearing range**
  - **Quieter sounds** that are masked by louder ones
- This makes the file smaller **without a big drop in quality**
- MP3 is a **lossy** format:
  - Some original data is **permanently lost**
  - You **can't get the original file back**

MPEG-4 (MP4) - Multimedia file format

- MP4 is similar to MP3, but stores **more than just audio**
- It can store:
  - **Music**
  - **Videos**
  - **Photos**
  - **Animations**
- Commonly used for **streaming videos** over the internet
- MP4 uses **compression** to keep file size small

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- Maintains **high quality** without noticeable loss to the viewer

Images

- **Bitmap images can be compressed**, the file size and quality are reduced
- **JPEG** is the most common file format for bitmap images
- The JPEG **compression algorithm creates a new file**, so the original file can no longer be used
- **Vector** graphics can also be compressed but **with limited success**
- **Scalable vector graphics** (.svg) use an XML text file which means that allows them to be compressed.

Run-length encoding (RLE)

What is run-length encoding?

- Run-length encoding (**RLE**) is a form of data compression that **condenses identical elements into a single value** with a count

Text data

- For text data containing the string "**AAAABBBCCDAA**", the plain RLE encoding would be "**4A3B2C1D2A**"
- The string has:
  - four 'A's (**4A**)
  - three 'B's (**3B**)
  - two 'C's (**2C**)
  - one 'D' (**1D**)
  - two 'A's (**2A**)
- To represent this in binary, the **count is stored in a fixed size binary format** (e.g. 7 or 8 bits)
- The character is stored using its **ASCII** value (7 bits)
- The binary RLE representation of **4A** would be **0000100 1000001**
  - **0000100** - binary for the count (4)
  - **1000001** - binary for 'A' (65)

Images

- Run-length encoding (**RLE**) uses **frequency/data pairs** to compress **bitmap image** data
- For example, the following bitmap image with a **bit depth of 1 bit** would have the following binary bit pattern

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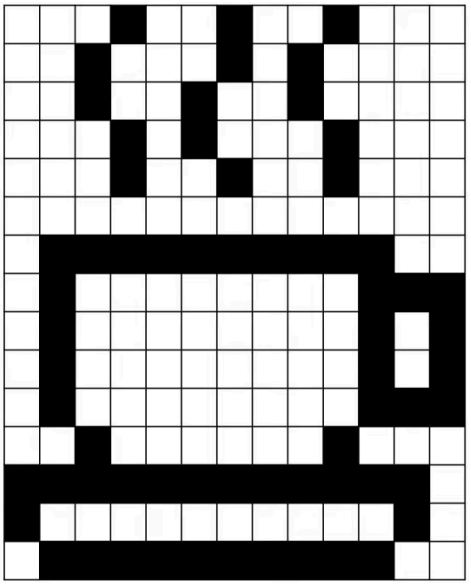
Bitmap	Bit pattern
	

0	0	0	1	0	0	1	0	0	1	0	0	0
0	0	1	0	0	0	1	0	1	0	0	0	0
0	0	1	0	0	1	0	0	1	0	0	0	0
0	0	0	1	0	1	0	0	0	1	0	0	0
0	0	0	1	0	0	1	0	0	1	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	0	0
0	1	0	0	0	0	0	0	0	0	1	1	1
0	1	0	0	0	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	0	0	1	1	1
0	0	1	0	0	0	0	0	0	1	0	0	0
1	1	1	1	1	1	1	1	1	1	1	1	0
1	0	0	0	0	0	0	0	0	0	0	1	0
0	1	1	1	1	1	1	1	1	1	1	0	0

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- Using RLE we **group pixel colours** and can create frequency/data pairs as follows
  - 3**0**, 1**1**, 2**0**, 1**1**, 2**0**, 1**1**, 5**0**, 1**1**, 3**0**, 1**1**, 1**0**, 1**1**, 6**0**, 1**1**.....

3x0 1x1 2x0 1x1 2x0 1x1

0	0	0	1	0	0	1	0	0	1	0	0	0
0	0	1	0	0	0	1	0	1	0	0	0	0
0	0	1	0	0	1	0	0	1	0	0	0	0

- **3 x 0** = 3 x white, **1 x 1** = 1 x black
- Data pairs **can carry over on to the next line**, e.g. end of first line and start of second line is 5 x 0 (**5 x white**)
- The **original** file would need to use **195 bits** of storage space
- The **RLE compressed** file would need use **38 bits** of storage space, **a saving of 157 bits (80%)**
- For images using more **than two colours**, the RGB value for each colour is stored along with the count
- For example, an image using the following four colours:

Colour	Red	Green	Blue
Black	0	0	0
White	255	255	255
Red	255	0	0
Green	0	255	0

- The data stored could look like:
  - 10 0 0 0 5 255 0 0 3 0 255 0
- **Ten black pixels** ((10) 0 0 0) - count + RGB
- **Five red pixels** ((5) 255 0 0) - count + RGB
- **Three green pixels** ((3) 0 255 0) - count + RGB

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